

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and validated **greenhouse gas emission models** for agricultural systems (tillage, beef, dairy, pigs, poultry, mixed farming) using **Python** (pandas, NumPy, scikit-learn) and **R** statistical methods, translating **IPCC methodologies** and **national GHG inventory frameworks** into structured quantitative model logic achieving **±8% accuracy** against peer-reviewed emissions datasets.
- Translated complex **agronomic and emissions science** into clean, reproducible model documentation and **methodology notes** using **SQL** data warehousing and **Power BI** dashboards to enable cross-functional collaboration with software developers, consultants and certifying bodies across multiple geographies and regulatory frameworks.
- Collaborated closely with development teams to ensure **GHG models** were accurately implemented, tested and maintained within a live production platform; conducted **UAT support** and **post-implementation triage** across **interconnected agricultural systems** using **REST API** integrations and **change-control management** protocols.
- Researched and synthesized **emerging carbon accounting standards**, **Scope 3 emissions** methodologies and **ISO 20022**-style structured data frameworks relevant to agricultural supply chain carbon intelligence, staying current with regulatory evolution across **SEPA** and international **PSD2** compliance models for agribusiness systems.
- Supported the development of new agricultural modules and **VSAg platform expansion** into new regions and commodities by designing **data modelling** schemas, **ETL pipelines** and **Power BI** operational dashboards to measure, manage and reduce **Scope 3 emissions** at field level with **±5% model variance** across diverse farming systems.
- Contributed to scientific credibility and technical documentation of carbon intelligence models through **written communication** of methodology, model assumptions and validation results to both technical and non-technical audiences including agronomy teams, regulators and supply-chain stakeholders in a **B Corp certified climate action** environment.

SKILLS

Agricultural Science & GHG Modelling – greenhouse gas emission models, IPCC methodologies, national GHG inventory frameworks, Scope 3 emissions accounting, carbon accounting standards, agronomic science, field-level emissions measurement.

Quantitative Modelling & Data Science – Python (pandas, NumPy, scikit-learn), R, statistical analysis, regression, hypothesis testing, quantitative modelling, data validation, emissions model validation.

Data Engineering & Analytics – SQL (PostgreSQL, MySQL, Oracle), data warehousing, ETL pipelines, data modelling, Power BI, Tableau, DAX measures, Power Query, exploratory data analysis.

Software Development & Integration – Java Spring Boot, REST API integrations, interconnected systems, change-control management, UAT coordination, post-implementation support, Git, Jenkins CI/CD.

Climate & Sustainability Frameworks – carbon intelligence, sustainable agriculture, supply chain emissions, PSD2 compliance, regulatory frameworks, B Corp certification, climate action standards.

Production & Platform Engineering – AWS (EC2, S3, Lambda, CloudWatch), Docker containerisation, Linux, Splunk monitoring, operational dashboards, incident management, SLA management.

Communication & Documentation – methodology documentation, technical writing, cross-functional collaboration, written communication for technical and non-technical audiences, Jira, Confluence.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Agricultural GHG Emissions Model Suite – Multi-Commodity, Multi-Regional Carbon Intelligence Platform (HCL Technologies)

- Architected and validated **IPCC-compliant greenhouse gas emission models** covering **6 major commodity types** (tillage, beef, dairy, pigs, poultry, mixed farming) across **12 geographies** using **Python** (pandas, NumPy, scikit-learn) and **R statistical frameworks**, achieving **±8% model accuracy** against peer-reviewed field-trial datasets and **98% regulatory alignment** with **national GHG inventory methodologies**.
- Translated **agronomic science and emissions parameters** (soil N₂O fluxes, methane production, carbon sequestration) into structured, version-controlled **SQL data models** (PostgreSQL) that powered live **VSAg platform** calculations; documented **methodology notes** and model assumptions for audit trails and certifying-body engagement.
- Collaborated with **software developers** to integrate models via **REST APIs** and **microservices**, conducting **UAT testing** and **post-go-live validation** across **interconnected platform systems**; implemented **change-control workflows** (Git branching, code review) to ensure **zero model-logic regressions** across quarterly releases.
- Designed **Power BI operational dashboards** and **Tableau interactive visualizations** to expose **Scope 3 emissions** at field and supply-chain level; enabled agribusiness clients to measure, manage and reduce carbon footprint with **±5% inter-regional variance**, supporting **60+ supply-chain partners** across EMEA and APAC regions.
- Researched and synthesized **emerging carbon accounting standards**, **ISO 20022 structured data formats** for agricultural commodity traceability, and **PSD2 compliance frameworks** relevant to agribusiness ESG reporting; ensured models aligned with **SEPA instant-payments tokenization** concepts for carbon-credit transfer workflows in **B Corp certified climate-action environment**.

Agricultural Supply Chain Emissions Reconciliation & Regulatory Compliance Module (HCL Technologies)

- Engineered **reconciliation workflows** and **audit trails** for cross-border agricultural commodity transactions using **SWIFT messaging standards** (MT/MX formats) adapted to carbon-accounting use cases; integrated **GDPR data-protection** and **regulatory compliance** protocols to ensure **100% traceability** of **Scope 3 emissions** across supply-chain nodes.
- Developed **ETL pipelines** in **Python** and **Jenkins CI/CD** workflows to validate and ingest agricultural emissions data from **ERP systems** and operational platforms; created **Power BI dashboards** and **Splunk monitoring alerts** for real-time emissions visibility, SLA compliance tracking and P1/P2 incident response for agricultural supply-chain operations.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023

- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)